Big Data Computing

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Cloud computing

1 | Overview

1.1 Basics

1.2 Setting up a virtual machine in the bwCloud

1.3 Running RStudio Server in the bwCloud

1.4 Running a Shiny Server in the bwCloud

1.5 Running a PostgreSQL database in the bwCloud

1.6 Running `Python` web-apps in the bwCloud

1 | Where are we?

Solutions	Problems: *Limited RAM*	Problems: *Processing power / time*
Local	▪ Code profiling ▪ Code optimization ▪ Optimized packages for data science	▪ Parallel computing
External	▪ Memory-efficient storage and computing ▪ Intro to Linux/shell/remote computing ▪ **Cloud computing** ▪ Cluster computing ▪ Big data frameworks	▪ Parallel computing on cluster/cloud

Basics

1.1 | If our machine is not powerful enough, we can *scale* to external resources!

DataCamp's definition of cloud computing:

> *"Cloud computing is the delivery of technology services - including compute, storage, databases, networking, software, and many more - over the internet with pay-as-you-go pricing."*

Using a cloud platform allows us to access better hardware and to potentially increase…

- the number of cores
- the available RAM
- the disk space
- …

Why use a cloud platform, and not simply buy the hardware yourself?

- Large entry cost
- Deployment and ongoing maintenance costs
- No ability to scale up quickly at need

Armbrust, Michael et al. (2009). Above the Clouds: A Berkeley View of Cloud Computing.
https://campus.datacamp.com/courses/cloud-computing-for-everyone/ [retrieved 12.06.2025]

1.1 | Forms of cloud services

Infrastructure-as-a-Service (IaaS)

- Basic hardware infrastructure, can be connected to and set up remotely
- Examples: Amazon Web Services, Microsoft Azure, Google Cloud, bwCloud, …

Platform-as-a-Service (PaaS)

- Hardware and software already (partially) set up, can be used to develop applications
- Examples: AWS Elastic Beanstalk, Microsoft Azure Web Apps, Google App Engine, bwUniCluster, …

Software-as-a-Service (SaaS)

- Software available over the internet
- Examples: Microsoft Office 365, Dropbox, Google Drive, Slack, Zoom, …

https://campus.datacamp.com/courses/cloud-computing-for-everyone/ [retrieved 12.06.2025]

1.1 | Major cloud computing providers (1/2)

Amazon Web Services (AWS)

- Amazon Elastic Compute Cloud (EC2)
  - → Instances / virtual machines
  - → *Elastic* = scalable & on-demand billing
- Amazon Simple Storage Service (S3)
- Amazon Relational Database Service (RDS)
- Amazon Elastic MapReduce (EMR)
  - → Distributed big data processing
  - → Contains managed Hadoop framework
- Amazon Redshift
  - → SQL-based data warehouse
- Kinesis, SageMaker, …

1.1 | Major cloud computing providers (2/2)

Microsoft Azure

- Azure Virtual Machines
- Azure Blob Storage
- Azure SQL Database
- Azure HDInsight
  - → Managed Hadoop distribution
  - → Supports open-source frameworks: Hadoop, Spark, Hive, Kafka, …
- Data Lake Storage, Stream Analytics, Machine Learning, …

Google Cloud Platform

- Google Compute Engine
- Google Cloud Storage
- Google Cloud SQL
- Big Query
  - → Fast data warehouse
  - → Allows standard SQL querying and analytics
- Cloud Bigtable, Cloud Datastore, Dataflow, AutoML, …

1.1 | Pros & cons of the major cloud computing providers

Benefits

- High fault-tolerance
- Automated backups
- Support
- Compatibility
- Services of one platform are intgerated
  - → Easy to connect different services (compute, storage, database, etc.)
  - → Single sign-on (SSO): login once and access all services seamlessly

Drawbacks

- Vendor lock-in: difficult to migrate between providers
- Cost complexity: pricing can be unpredictable and complex
- …
- **We have to pay** (at least for everything that goes beyond a trial version)

→ The bwCloud is currently free of charge for us, that's why we will use it in this course!

→ Setup works very similar on other cloud platforms

1.1 | The bwCloud is an IaaS for science and education

- The bwCloud is an IaaS environment
- You can create, start and operate virtual machines (VMs)
  - → Choose CPUs, RAM, and storage space
  - → VMs are very quickly set up and shut down
  - → Scalable and flexible: computing power and/or storage capacity can be upgraded
- All bwCloud machines are managed from the OpenStack dashboard
- The bwCloud expects the following previous knowledge:
  - → What is SSH?
  - → How can a public SSH key be generated?
  - → What is a (virtual) server?
  - → How do I install software on my server and how do I keep the system up-to-date?

→ You need to be able to independently operate your VM, the documentation is relatively scarce!

https://www.bw-cloud.org/en/first_steps [retrieved 12.06.2025]

1.1 | The bwCloud is built upon OpenStack

Free, open source cloud computing platform

- Provides IaaS functionality
- Controls large pools of processing, storage and network resources throughout a datacenter
- Can be controlled via an intuitive dashboard interface

Example users

- Deutsche Telekom (Open Telekom Cloud)
- Wikimedia Labs
- CERN
- Schwarz-Gruppe (Lidl, Kaufland)
- Many more…

https://www.openstack.org/software/ [retrieved 12.06.2025]

1.1 | The main advantage of cloud computing is *scalability*

Scalability in cloud computing: the ability to add hardware in order to solve complex tasks

- **Vertical** scalability: increase the resources (CPUs, RAM, hard disk, …) of a single machine
- **Horizontal** scalability: increase the number of machines/nodes connected to a cluster

On a single machine in the bwCloud, these flavors are available:

Name	# vCPUs	RAM (in GB)	Root disk (in GB)
m1.nano	1	0,5	12
m1.tiny	1	1	12
m1.small	1	2	12
m1.medium	2	4	12
m1.large	4	8	12
m1.xlarge	8	16	12
m1.xxlarge	16	32	12
win.medium	2	4	60
win.large	4	8	60
win.xxlarge	16	32	60

https://www.bw-cloud.org/de/bwcloud_scope/flavors [retrieved 12.06.2025]

1.1 | Scaling in the bwCloud

- Default quota after registration:
  - → 1 Instance
  - → 1 vCPU
  - → 1 GB RAM
  - → …
- You can request a larger quota (see here)
  - → **Careful**: there very likely will be a billing system in the future!
- If you have sufficient quota, you can scale up/down your machine under: *Instances* → column *Actions* → *Resize Instance*

Using volumes to increase disk space

- The root disk by default is very limited (12 GB)
- *Volumes* are used to increase storage (see *Volumes* in dashboard)
  - → Create volume with the desired size
  - → Attach the volume to the instance you want to use it for
  - → The *Attached To* column shows the path where the storage is located
  - → See slide 27 for further procedure

https://www.bw-cloud.org/en/faq/quota [retrieved 12.06.2025]
https://www.bw-cloud.org/en/faq/instances [retrieved 12.06.2025]

1.1 | The costs of cloud computing

There are a lot of different pricing schemes, probably the most popular:
Usage-based pricing / Pay-as-you-go / Pay for the hour

- Pay per instance and hour/second
- Stop/terminate instance when no longer needed (rewards conservation)
- Turns potentially large fixed costs into smaller variable costs

Armbrust, Michael et al. (2009). Above the Clouds: A Berkeley View of Cloud Computing.

1.1 | The costs of cloud computing: example for AWS

On-Demand Plans for Amazon EC2

Select a location type and region

Location Type: AWS Region

Region: Europe (Frankfurt)

Select an operating system, instance type, and vCPU to view rates

Operating system: Linux

Instance type: General Purpose

vCPU: 4

Viewing 26 available instances

1 2

Instance name	On-Demand hourly rate	vCPU	Memory	Storage	Network performance
t4g.xlarge	$0.1536	4	16 GiB	EBS Only	Up to 5 Gigabit
t3.xlarge	$0.192	4	16 GiB	EBS Only	Up to 5 Gigabit
t3a.xlarge	$0.1728	4	16 GiB	EBS Only	Up to 5 Gigabit
t2.xlarge	$0.2144	4	16 GiB	EBS Only	Moderate

https://aws.amazon.com/ec2/pricing/on-demand/ [retrieved 12.06.2025]

1.1 | Costs of bwCloud (1/2)

When you request more quota on the bwCloud, you will get a (serious) e-mail:

Currently, the use of bwCloud resources is free of charge. However, we must inform you that we are currently working on a "price list" for the various flavors / quota settings, as we are aiming to charge for the services used in the future. This price list is not yet fixed and is in the process of coordination with all partners and parties involved. Therefore, we would like to use this e-mail to basically prepare you for the topic of "service charging" in order to avoid possible surprises. Should a service settlement occur, this will take place in a staged process with a sufficiently large lead time.

Specifically, this means:

1. *first, the price list will be communicated to all users of bwCloud.*
2. *a transition phase will then be set up, during which use will still be free of charge.*
3. *as of a key date X, billing will then begin. The transition phase will last a few months. There will be no retroactive billing for services that have already been used.*

***PLEASE NOTE. If employees of an institution (chair, data center etc.) without their own budget want to use bwCloud SCOPE, first get the consent of your supervisor or the person in charge. We want to prevent that no unresolved (financial) situations arise.*

*As a student you do *not* have access to a cost center. Therefore please contact your supervisor in any case. If you have taken note of this information and accept it, please let us know via Reply here and we will set up the project. Please follow our instructions as described under [1]. This will help us to avoid frequent queries. Please enter missing information.*

1.1 | Costs of bwCloud (2/2)

Main takeaways

- For now, using the bwCloud is free of cost
- However, they are working on a billing scheme for the future
- You will be informed once that billing scheme is launched
  - → You will then need the permission of a supervisor
- Be aware that anything you do on the bwCloud leads to costs for someone (e.g., the university)
  - → Use resources responsibly
  - → Stop a machine if you are no longer using it
  - → In other cloud computing environments, you would directly be billed for every hour your machine is running!

1.1 | When to use the bwCloud

Limitations

- The bwCloud may be not ideal to work with truly **big** data (hundreds of gigabytes or terabytes) because of limited read and write speeds for the disks/the network:
  - → Typical read/write for HDDs: 80-160 MB/s
  - → Typical read/write for SSDs: 200-550 MB/s
  - → On bwCloud: throttled to 100 MB/s or 800 IOPS
- The processors are not the fastest

Typical use cases

- Hosting a website/web-app
- Hosting a database (not for very large datasets, see above)
- Hosting an API
- Tasks that run frequently but are computationally not very expensive, e.g., scraping a website regularly

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https://www.bw-cloud.org/en/faq/storage [retrieved 12.06.2025]
https://tekie.com/blog/hardware/ssd-vs-hdd-speed-lifespan-and-reliability/ [retrieved 12.06.2025]

Setting up a virtual machine in the bwCloud

1.2 | Registration on the bwCloud

1. Go to `https://portal.bw-cloud.org`
2. Choose *bwIDM via OpenID Connect* as authentication
3. Select your home organization ("Universität Tübingen") and login with your standard university credentials
4. Register for bwCloud SCOPE by agreeing to the terms of use
5. You are automatically redirected to the bwCloud dashboard and can start with the setup process

https://www.bw-cloud.org/en/first_steps [retrieved 12.06.2025]

1.2 | Create SSH keys for remote access

- If you have a terminal/shell, you can directly use `ssh-keygen -t rsa -b 4096`
  - → By default, the public key is stored in `id_rsa.pub`, the private key in `id_rsa`
- If not / alternatively, you can either use PuTTYgen or MobaKeyGen.
  1. Download and start the program:
     - PuTTYgen: start puttygen.exe
     - MobaXterm: start MobaXterm, click *Tools* → *MobaKeyGen (SSH key generator)*
  2. Click *Generate* and follow the instructions
  3. Copy the displayed public key into a simple `.txt` file and save the file
  4. Click *Save private key* and save the private key in the same directory

→See chapter 7 (Linux shell) for more detailed information

https://www.bw-cloud.org/en/bwcloud_scope/use [retrieved 12.06.2025]

1.2 | Register your SSH key in the bwCloud

1. Login at `https://portal.bw-cloud.org`
2. Go to *Project* → *Compute* → *Key Pairs*
3. Click *Import Public Key*
4. Give your SSH key a unique name
5. Select *SSH Key* as *Key Type*
6. Copy the full public SSH key that you previously saved into the text field
7. Click *Import Public Key*
8. You should now see the imported key in the overview

https://www.bw-cloud.org/en/bwcloud_scope/use [retrieved 12.06.2025]

1.2 | Starting your own instance (VM) in the bwCloud

1. Login at `https://portal.bw-cloud.org`
2. Go to *Project* → *Compute* → *Instances*
3. Click *Launch Instance*
4. Give your instance / virtual machine a meaningful name
5. Under *Source*, choose the image/distribution which you want to base your instance on
   - → Select an image by clicking on the arrow up on the right (I would recommend Ubuntu 24.04 for now; if you are experienced on Linux, feel free to choose another one)
   - → Discard the selection of an image by clicking on the arrow down
6. Under *Flavor*, select the resources needed (again by arrow up/down)
   - → A yellow warning symbol indicates that your quota is too low for that flavor
   - → Recommendation: choose *m1.large*, if your quota allows it, if not, *m1.tiny* is fine
7. Under *Key Pair*, select the key pair you want to have associated with this instance
   - → This integrates the public SSH key into the instance
   - → Later, you will access that instance by using the corresponding private key
8. Usually, you can leave all other settings at their default
9. Click *Launch Instance* to start the instance (might take a couple of seconds)

https://www.bw-cloud.org/en/bwcloud_scope/use [retrieved 12.06.2025]

1.2 | Set up the SSH client (or connection)

This assumes you use PuTTY (very similar to MobaXterm or other SSH clients)

1. Start PuTTY
2. Go to *Connection* → *SSH* → *Auth*
3. Under *Private key file for authentication:* click *Browse* and open your private SSH key
4. Optional, but useful: save current PuTTY settings (private SSH key, Host Name, …)
   - → Go back to *Session*
   - → Enter a name for the session under *Saved Sessions*
   - → Click *Save*
   - → The next time you open PuTTY, select the name of the saved session and click *Load*

If you have a terminal/shell on your machine and don't want to use an SSH client, you can also directly use the `ssh` command: `ssh <username>@<remote_host>`

- By default looks for the private key in `~/.ssh/id_rsa` (alternatively, specify location with `-i` flag)
- For more convenience, you can create a `config` file that stores the credentials (IPs, usernames, keys, …) for different remote connections

https://www.bw-cloud.org/en/bwcloud_scope/use [retrieved 12.06.2025]

1.2 | Connect to your instance using SSH

1. Open your SSH client (e.g., PuTTY)
2. Configure the settings: private SSH key, IP address, …
   - → If previously saved, load the Session that contains the specific settings
   - → If not, follow the instructions on the previous slide
   - → Find the IP address of your instance: in the bwCloud overview of your instances (*Project* → *Compute* → *Instances*), the first line under *IP Address*
   - → Copy and paste the IP address into the PuTTY field *Host Name (or IP address)*
3. Click *Open* or if you use `ssh` directly, type the full command in your terminal
4. A new shell opens and asks for your login name (only if not specified in `ssh`)
   - → Look up the standard SSH username for your image/distribution here
   - → Enter the standard SSH username (e.g., `ubuntu`)
5. After successful authentication, you have access to your virtual machine and can start working with it!
   - → Remember, this is a clean, newly set up machine without a lot of pre-installed software
   - → But it is **your** machine, you have all privileges to install and modify!

1.2 | Mounting volumes to your VM (1/2)

1. Create a volume and attach it to the instance (see slide 14)
2. Find the device path with

```
lsblk
```

3. Format the volume on the instance:

```
sudo mkfs.ext4 <path_to_volume>  # E.g., sudo mkfs.ext4 /dev/vdb
```

4. Create a directory where you want to access your volume ("mount point")

```
mkdir <path_to__mount_point>  # E.g., /home/ubuntu/data
```

https://www.bw-cloud.org/en/faq/storage#faq_storage_3 [retrieved 12.06.2025]

1.2 | Mounting volumes to your VM (2/2)

5. Add this mount point to `/etc/fstab`
   - → Leads to the volume being mounted automatically on startup
   - → Using `nano`, add a new entry (new line) at the end of the file:

```
sudo nano /etc/fstab
```

```
# Add this line at the end of the file:
<path_to_volume>     <path_to_mount_point>    ext4    defaults    0 0
# E.g., /dev/vdb    /home/ubuntu/data    ext4    defaults    0 0
```

6. Mount it manually for now (otherwise reboot necessary):

```
sudo mount <path_to__mount_point>  # E.g., /home/ubuntu/data
```

7. Change the ownership as needed, e.g.,

```
sudo chown -R <user_name> <path_to_mount_point>  # E.g., sudo chown -R ubuntu /home/ubuntu/data
```

8. Verify the available storage space with `df -h`

https://www.bw-cloud.org/en/faq/storage#faq_storage_3 [retrieved 12.06.2025]

1.2 | Increasing the size of a volume

1. Extend the volume in the OpenStack dashboard
   → *Volumes* → column *Actions* → *Extend Volume*
2. Reboot the instance in the OpenStack dashboard
   → *Instances* → column *Actions* → *Hard Reboot Instance*
   → This will restart your engine, all temporary (not saved) data will be lost
3. Extend the file system

```
sudo resize2fs /dev/vdb  # Replace with actual path to volume
```

Note: it is only possible to increase the size of a volume. If you want to decrease the size, you have to create a second (smaller) volume and transfer the data to this new volume, then delete the old one.

1.2 | Network & security groups

- By default, a new VM in the bwCloud is initially only accessible from outside via SSH (port 22)
  - → This is defined in the assigned security group, e.g. *default*
  - → All other ports are closed
- Either create a new security group or modify an existing one
  - → As soon as the rules of a security group change, these changes take effect for all associated instances.
  - → No reboot required

Example use cases

- You are running a website that should be available via HTTPS (port 443)
- You are running a database
- You are running a Shiny Server
- You are running a RStudio Server
- …

→ The respective port must be open in one of the assigned security groups

https://www.bw-cloud.org/en/bwcloud_scope/use#open_port [retrieved 12.06.2025]

1.2 | Open a port for your VM

- In the OpenStack dashboard go to *Network* → *Security Groups*
- Either create a new security group or modify an existing one
- *Manage Rules* → *Add Rule*:
  - → Rule: either choose a preset (e.g., HTTPS) or choose *Custom TCP Rule*
  - → Direction: define the direction, i.e. *Ingress* for incoming connections or *Egress* for outgoing connections (for the example applications in these slides you only need *Ingress*)
  - → Open Port: *Port*
  - → Port: the port you want to open, e.g. 5433
  - → Remote: *CIDR*
  - → CIDR: 0.0.0.0/0 (or specify a limited IP range, e.g., your organization - for the University's network that would be 134.2.0.0/16)

https://www.bw-cloud.org/en/bwcloud_scope/use#open_port [retrieved 12.06.2025]

1.2 | Now you have your Linux server running

- This is just a plain Linux machine
  - → You can do anything we learned in the previous chapter
  - → There are plenty of resources in the internet
- The next sections provide some guidance how to install and run a selection of potentially useful software on your server
  - → Apart from that, there are great tutorials in the internet, in case your desired software is not covered in these slides

The first thing you should do: update all software!

```
sudo apt-get update
sudo apt-get upgrade
```

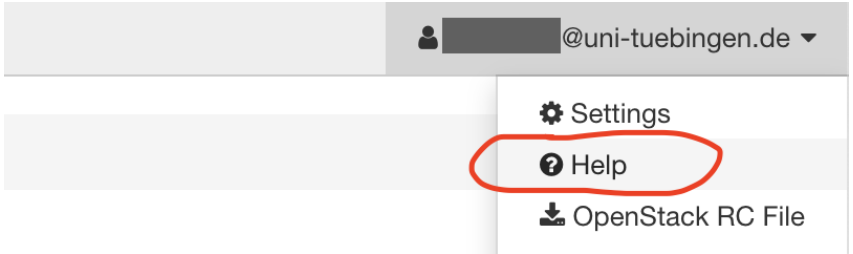
1.2 | Getting help

Software related

- Examples: error when installing software, questions about configuration files, …
- Since your VM is just a plain Linux machine, consult the internet resources or your favorite coding assistant for help

Hardware related

- Examples: VM not reachable with SSH, mounted volumes not working properly, need more quota, …
- The bwCloud has an excellent support
  → Click the top-right corner in the OpenStack dashboard to access the BW support portal



Running RStudio Server in the bwCloud

1.3 | A browser-based interface to R running on a remote Linux server

"RStudio Server enables you to provide a browser based interface to a version of R running on a remote Linux server, bringing the power and productivity of the RStudio IDE to server-based deployments of R."

- You can access the RStudio IDE in a way very similar to your local machine
- Differences:
  - → Scalable resources, potentially very large
  - → Running on Linux: some installations may be different
  - → Multiple users can access the same machine from different locations

→ The next slides show how to set up RStudio Server on an Ubuntu VM in the bwCloud

https://rstudio.com/products/rstudio/download-server/ [retrieved 12.06.2025]

1.3 | Setting up the VM for RStudio Server installation

This assumes you have a VM with an Ubuntu image running on the bwCloud

1. Install R (if not already done)

```
sudo apt-get install r-base  # Ideally, also install r-base-dev for installations from source
```

2. Install dependencies for R packages:

```
sudo apt-get -y install libcurl4-gnutls-dev libxml2-dev libssl-dev
```

https://cran.r-project.org/bin/linux/ubuntu/fullREADME.html [retrieved 12.06.2025]

1.3 | Downloading and installing RStudio Server itself

1. Install gdebi, a tool that is used to install RStudio Server and all its dependencies:

```
sudo apt-get install gdebi-core
```

2. Download and install RStudio Server (check for latest version!):

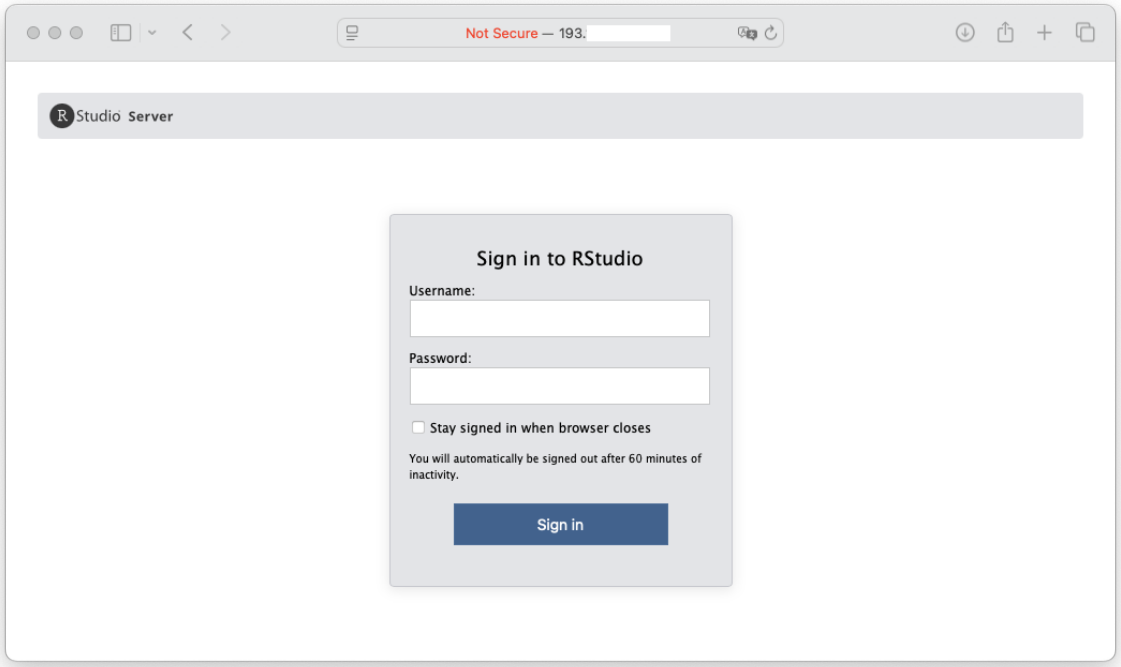
```
wget https://download2.rstudio.org/server/jammy/amd64/rstudio-server-2025.05.1-513-amd64.deb
sudo gdebi rstudio-server-2025.05.1-513-amd64.deb
```

3. By default, RStudio Server can be accessed at port 8787

- The browser interface is available at `http://my_ip_or_server:8787` (where `my_ip_or_server` is either the IP address of the instance or a web domain pointing to that IP)
- Access to that port has to be granted from the bwCloud dashboard (see slide 31)

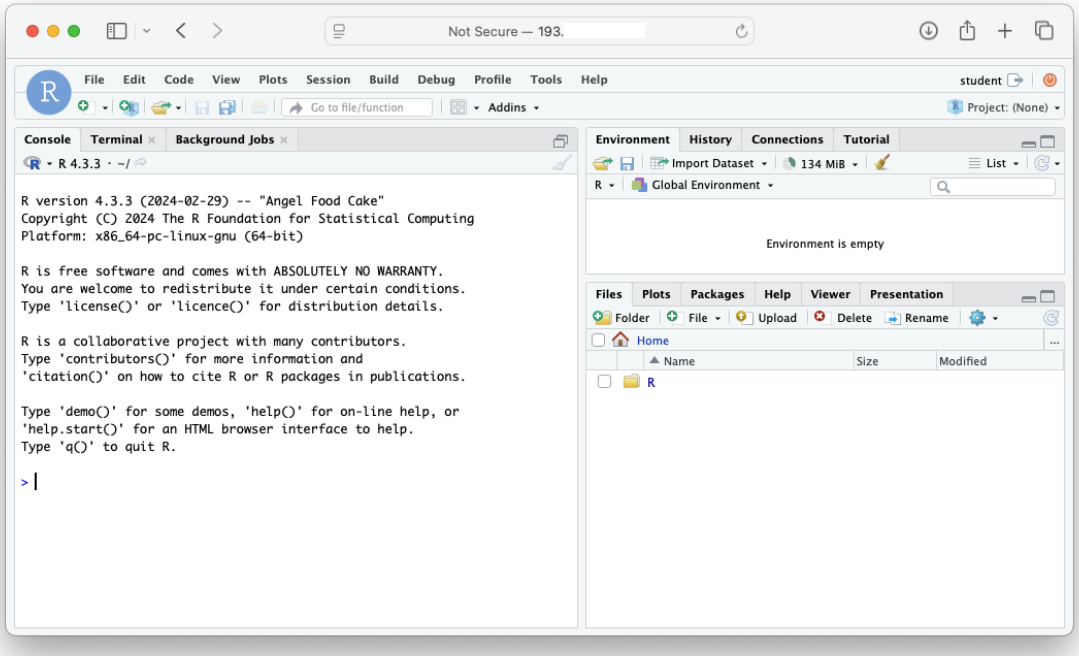
https://rstudio.com/products/rstudio/download-server/ [retrieved 12.06.2025]

1.3 | RStudio Server can be accessed with username and password



- By default, RStudio Server uses the usernames and password of the Linux server
- See chapter 7 (*Intro to Linux*) for more details on user and permissions management

1.3 | After logging in, you see the familiar RStudio interface



- You can install packages from here or directly from the shell
  - → By default, packages are only installed for the current user
- Transfer files using any FTP solution, e.g., FileZilla
- RStudio Server accesses the scalable resources of the VM

Running a Shiny Server in the bwCloud

1.4 | Why use your own server for hosting Shiny apps?

- Free accounts on shinyapps.io are very limited in size, number of accesses, and time until timeout
- Hosting your own server gives you full control over the app
- You can host your app in a controlled environment (e.g., within you organization), such that the app and the data never leaves a secure network
- You can also make the apps accessible across the internet

Shiny Server

- Free open source back end program for Shiny web applications
- Lets you host multiple Shiny apps on a single server
- Available for Ubuntu, Red Hat/CentOS, and openSUSE
- Is only the backend, one needs a server to host the app
- Can also host (interactive) R Markdown documents

→ The next slides show how to set up Shiny Server on an Ubuntu machine in the bwCloud

https://shiny.rstudio.com/articles/shiny-server.html [retrieved 12.06.2025]

1.4 | Installing Shiny Server

This assumes you have a VM with an Ubuntu image running on the bwCloud

1. Install R, see slide 36, if not already done
2. Install the `{shiny}` package before installing the Shiny Server

```
sudo su - -c "R -e \"install.packages('shiny', repos='https://cran.rstudio.com/')\""
```

3. Install gdebi (tool that is used to install Shiny Server and all its dependencies), if not already done

```
sudo apt-get install gdebi-core
```

4. Download and install Shiny Server (check for latest version**!**):

```
wget https://download3.rstudio.org/ubuntu-20.04/x86_64/shiny-server-1.5.23.1030-amd64.deb
sudo gdebi shiny-server-1.5.23.1030-amd64.deb
```

https://posit.co/download/shiny-server/ [retrieved 12.06.2025]

1.4 | Accessing Shiny Server remotely

- Once Shiny Server is installed, there is a directory `/srv/shiny-server`
  - → Any applications and assets stored here will be made available by Shiny Server
- By default, Shiny Server can be accessed at port 3838
  - → A Shiny app stored in `/srv/shiny-server/myApp` will be available at `http://my_ip_or_server:3838/myApp` (where `my_ip_or_server` is either the IP address of the instance or a web domain pointing to that IP)
  - → Access to that port has to be granted from the bwCloud dashboard (see slide 31)
- You can test the Shiny Server with pre-installed example apps
  - → `http://my_ip_or_server:3838` should show a welcome page
  - → In the `/srv/shiny-server/sample-apps` directory, there are two example apps which can be accessed by, e.g., `http://my_ip_or_server:3838/sample-apps/hello`

https://shiny.rstudio.com/articles/shiny-server.html [retrieved 12.06.2025]

1.4 | Upload your own apps to Shiny Server

- Shiny apps have to be uploaded as two separate files, one UI file called `ui.R`, and one server file called `server.R`
- Both files have to be stored in the same directory inside `/srv/shiny-server/`
  - → You can create subdirectories, e.g., `/srv/shiny-server/my-apps/nice-app`
  - → In the url to your app, specify this path (relative to `/srv/shiny-server`), e.g., `http://my_ip_or_server:3838/my-apps/nice-app`
- Loading files to `/srv/shiny-server/` needs permissions that the `ubuntu` user does not have by default. Two options:
  - → Change permissions
  - → Store the files elsewhere and link to the shiny server directory

Running a PostgreSQL database in the bwCloud

1.5 | Install PostgreSQL

This assumes you have a VM with an Ubuntu image running on the bwCloud

1. Install PostgreSQL

```
sudo apt update
sudo apt install postgresql postgresql-contrib
```

2. Start and enable PostgreSQL service

```
sudo systemctl start postgresql
sudo systemctl enable postgresql
```

3. Check the status to confirm PostgreSQL is running

```
service postgresql status
```

1.5 | Set up PostgreSQL for remote connections (1/3)

This is only relevant, if you want to access your database from another machine (usually the case)

Allow remote connections

1. Edit the configuration file
   → Uncomment and edit the listen_addresses attribute to start listening to all available IP addresses

```
sudo nano /etc/postgresql/14/main/postgresql.conf  # Adjust for PostgreSQL version (here: 14)
```

```
listen_addresses = '*'
```

2. Edit the PostgreSQL access policy configuration file
   → Append a new connection policy at the bottom of the file
   → The pattern is as follows: `[CONNECTION_TYPE] [DATABASE] [USER] [ADDRESS] [METHOD]`

```
sudo nano /etc/postgresql/14/main/pg_hba.conf  # Adjust for PostgreSQL version (here: 14)
```

```
# Add this line at the end of the file (adjust if you want to restrict access)
host all all 0.0.0.0/0 md5
```

1.5 | Set up PostgreSQL for remote connections (2/3)

2. Edit the PostgreSQL access policy configuration file (cont'd)
   - → We are allowing TCP/IP connections (`host`) to all databases (`all`) for all users (`all`) with any IPv4 address (`0.0.0.0/0`) using an MD5 encrypted password for authentication (`md5`).
   - → If you want a different configuration (e.g. limited range of IP addresses) adjust it accordingly

Change the port

The default port of PostgreSQL is 5432 which is reserved for something else in the bwCloud. Therefore, we have to change the port for PostgreSQL, e.g. to 5433.

1. Edit the configuration file
   - → Change the port number from 5432 to e.g. 5433

```
sudo nano /etc/postgresql/14/main/postgresql.conf  # Adjust for PostgreSQL version (here: 14)
```

```
port = 5433  # Change this line
```

2. Grant access to that port from the bwCloud dashboard (see slide 31)

1.5 | Set up PostgreSQL for remote connections (3/3)

Restart and check

1. Restart the PostgreSQL service to apply the changes

```
sudo systemctl restart postgresql
```

2. Make sure the system is listening to the port that is reserved for PostgreSQL

```
ss -nlt | grep 5433
```

1.5 | Manage your databases

- With `psql` you can manage your PostgreSQL databases from the command line:

```
sudo -u postgres psql
```

- This will open a `psql` terminal where you can…
  - → set a password for the `postgres` user (admin user with all privileges, should not be shared with someone else)
  - → create new users and set their passwords
  - → create new databases
  - → manage connections
  - → run SQL queries
  - → …
- Some useful commands:
  - → Exit: `\q` or `exit`
  - → List of all the databases that are available on a server: `\l`
  - → List of all the users with their privileges: `\du`
  - → Connect to a database: `\c database_name`

→ Search the internet, there are plenty of tutorials on how to use PostgreSQL and `psql`

Running `Python` web-apps in the bwCloud

1.6 | Running `Python` web-apps in the bwCloud

Django

- Full-featured web framework with built-in admin, ORM, and security features
- Check out tutorials in the internet, e.g. this one

Flask

- Lightweight, flexible micro-framework for simpler web applications
- Check out tutorials in the internet, e.g. this one

Setup

- To run both frameworks in production, a common way is to use `gunicorn` to run the app and `nginx` as a reverse proxy (the linked tutorials show how to)
- Check the security groups in the bwCloud dashboard if all required ports are open

Questions?